Sahaj Bhadja

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Summary

Software engineer with 2+ years building web applications, operational workflows, and secure cloud services. Experienced in JavaScript, TypeScript, Java, Python, HTML, XML, SQL, REST APIs, debugging, testing, documentation, and deployment.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built Python APIs, Azure SQL services, and operational dashboards for 500+ users, translating user requirements into secure interfaces and indexed queries that reduced response latency from 1.2 seconds to 350 milliseconds.
- Implemented fault-tolerant workflows for 5,000+ monthly jobs using explicit states, retries, reconciliation, and failure isolation, tripling throughput while documenting controls, improving observability, and supporting reliable production operations.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed responsive React interfaces and Flask REST services for upload, routing, status, and history workflows, translating operational requirements into maintainable modules supporting 150+ users and 10,000+ daily requests.
- Created concurrent processing for 1,000+ files with automated program tests, input validation, AWS RDS persistence, and structured error handling, reducing turnaround 60% while maintaining stable system behavior.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built reproducible scientific simulation programs processing 100,000+ records, applying automated tests, instrumentation, input-output validation, and systematic debugging to improve long-horizon planning accuracy 18% across controlled system evaluations.
- Conducted cross-functional design reviews and validation experiments, documenting technical program modifications and tuning a spatial-temporal architecture that improved closed-loop system stability 15% across repeated controlled simulation scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed object-oriented web services and ETL integration modules using REST, AWS Glue, S3, and SQL, improving response times 30% and workflow efficiency 68% across complex system interfaces.
- Maintained 20+ automated program and integration test suites with Jenkins, PyTest, and Mockito across Windows, Linux, and macOS, accelerating debugging, defect correction, and confidence in production upgrades.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular JavaScript, React, and Node.js interfaces across eight secure authentication, scheduling, permissions, and notification workflows, synchronizing 2 million+ Azure SQL records through maintainable services and WebSockets.
- Automated Docker builds, program tests, staged deployments, and production monitoring through Jenkins and Azure, reducing release time 50% while improving modification traceability, system maintenance, and operational reliability.

Skills

Languages & Web: JavaScript, TypeScript, Java, Python, C++, HTML5, CSS3, XML, React, Node.js, Flask, responsive interfaces
Applications & Data: REST APIs, JSON, WebSockets, client-server architecture, OOP, design patterns, API integration, workflow applications, SQL, Azure SQL, SQL Server, PostgreSQL, MySQL, MongoDB, AWS RDS
Cloud & Operations: Azure, AWS, S3, Blob Storage, Git, Jenkins, CI/CD, Docker, PyTest, Mockito, program testing, unit testing, system testing, integration testing, input-output validation, debugging, documentation, deployment, maintenance, observability, Agile

Projects

HMIS | Live | *TypeScript, JavaScript, React, REST APIs, Appwrite, Sentry*
- Built secure web workflows for appointments, records, and consent-based access using JavaScript interfaces, REST APIs, authentication, automated notifications, and Sentry monitoring, reducing administrative workload 30% while improving issue traceability.

Chat2SQL | Code | *Java, TypeScript, JavaScript, HTML, SQL, Docker*
- Built a cross-platform application supporting 10+ databases with Java services, JavaScript and HTML interfaces, schema-aware input validation, Docker deployment, and debugging workflows for reliable data management and integration.

CPU-scheduling Simulator | Live | *C++, React, JavaScript, EJ-2*
- Developed executable scheduling programs with interactive process inputs, automated calculations, edge-case validation, and Gantt-chart diagrams across six algorithms, reducing debugging time 20% through deterministic simulation and documented system behavior.